

State encoding

An FSM has abstract states S_0 , S_1 , S_2 , but silicon stores bit patterns in flip-flops

Hamming distance and flips

- Starter: four states in binary, two flip-flops
- Click S1 to S2 and Hamming distance is two, both bits flip
- That matters because combinational decode can briefly see illegal intermediate codes
- Switch to one-hot
- Switch to Gray: ring steps often have Hamming one
- Yellow rows in the arc table flag multi-bit transitions

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / State encoding

INTERACTIVE TOOL

State encoding lab

Compare binary, one-hot, and Gray encodings for the same FSM: flip-flop count, code table, and how many bits flip on each arc (Hamming distance — multi-bit flips risk decode glitches). Starter: 4 states in binary.

Starter example: 4 states, binary (2 FFs). Open One-hot and Gray tabs; click S1→S2 in binary to see Hamming distance 2.

Challenge 0 / 22 cleared

Quiz: binary FFs: Binary encoding of N states needs how many flip-flops?

☐ $\lceil \log_2 N \rceil$

☐ N

☐ N-1

☐ 2N

Show hint

Check

Next

Load challenge setup

Idle

Quiz: binary FFs

Quiz: one-hot FFs

Quiz: Gray

Quiz: Hamming

Quiz: glitch risk

Quiz: one-hot decode

Quiz: unused codes

Quiz: tradeoff

4-state binary

4-state one-hot

5-state binary FFs

8-state one-hot

State encoding starter

Workbook practice

- In the workbook track
- Sketch one-hot for three states
- Explain why five states in three binary bits leaves unused codes
- Name one pitfall: multi-bit flips glitching state decode

Pitfalls to watch

- Do not assume binary is always best, compact codes can hazard decode
- One-hot costs more FFs but simplifies “are we in S2?” checks
- Unused binary codes need a default next-state, not hope
- And remember: the browser lab is literacy
- Real RTL still needs synthesis encoding hints and safe illegal-state recovery

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, draw one binary multi-flip arc and one Gray single-flip step
- When you are ready, take the short quiz, then continue to the sequence detector